

Mohammad Haider

Software Engineer | ML experiments & research tooling

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SUMMARY

Software engineer with three years at Microsoft and subsequent BI development at the University of Virginia. Combines production data-system validation and internal-tool delivery with independent PyTorch language-model experiments, reproducible evaluation workflows, and simulation tooling.

PROFESSIONAL EXPERIENCE

University of Virginia | Charlottesville, VA

IT Analyst Intermediate, BI Developer

May 2025 - Mar 2026

- Delivered SQL-backed Tableau products from stakeholder interviews through deployment and maintenance, translating ambiguous operational questions into shared metric definitions.
- Reconciled purchase-order states across financial systems and built reporting that surfaced discrepancies and their financial impact.
- Collaborated on telemetry-informed optimization that reduced organization-wide dashboard load times by approximately 20%.

Microsoft | Redmond, WA

Software Engineer I

Jul 2021 - Jul 2024

- Owned an SSAS Multidimensional-to-Tabular migration; defined parity checks, monitored four weeks of pipeline behavior, and kept the legacy service available until cutover criteria were met.
- Wrote and deployed Azure Data Factory verification scripts for daily pipeline monitoring and legacy-to-Tabular data consistency.
- Validated downstream reporting workflows showing up to 200x faster filtering and approximately 4x faster Excel loading.
- Built a full-stack C#.NET prototype that guided configuration, generated configuration files, and validated output.

Microsoft | Redmond, WA

Software Engineering Intern

Summers 2017, 2018

- Built a C# service that identified each OneDrive user's most recently used items for a product UI feature; validated the service in development.

SELECTED INDEPENDENT PROJECTS

LM Lab | Python, PyTorch, Jupyter

github.com/haidmoham/lmlab

- Implemented byte-pair tokenization, causal attention, and transformer blocks in PyTorch; compared a bigram baseline with 1- to 8-block models.
- Ran controlled comparisons across three paired seeds with matched data exposure and update budgets; recorded held-out loss, parameter counts, and timing.
- Preserved source hashes and model/optimizer/RNG checkpoints, with restoration checks that reproduced subsequent updates.

C-1N + Robotics Test Bench | Python, MuJoCo

mhaider.dev/c1n

- Built a reproducible 10-second standing test and a 33-case disturbance suite for an 18-joint simulation, retaining recovery failures as explicit limits.
- Recorded shared simulation telemetry for headless runs and Jupyter diagnostics, separating measurement from interpretation.

TECHNICAL SKILLS

Languages: Python, C#, SQL, R

Production systems: C#.NET, Azure Data Factory, SSAS Tabular, SQL-backed analytics

Independent projects: PyTorch, NumPy, pandas, Jupyter, MuJoCo; Git

EDUCATION

Virginia Tech | B.S. in Statistics

2021